

Chemical Vapor Transport Reactions – A Historical Review

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In Memoriam of Professor Harald Schäfer on the Occasion of His 100th Anniversary

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Abstract. Since their first recognition in mineral forming processes some 150 years ago chemical vapor transport reactions (CVTR) have attracted continuous scientific interest. Due to the pioneering work of *Harald Schäfer* quantitative understanding and exploitation of transport reactions for crystal growth, synthesis, investigation of high-tem-

perature gas species, and thermodynamic studies have become possible. Renewed interest in CVT is triggered by the demand of material sciences for novel compounds with tailor-made physical properties and by the need for efficient recycling strategies for various metals from industrial waste.

1 Introduction

For solid state scientists the term “chemical vapor transport” refers to a synthetic method on laboratory scale for the preparation and purification of solid materials usually in form of single crystals. Up to now more than 2000 papers have been published on this topic in chemical, physical, crystallographic, and mineralogical journals. It is well worth noting that about

15% of these articles appeared in ZAAC. In these papers the crystallization of elements, intermetallics, halides, oxides, sulfides, selenides, tellurides, chalcogenide-halides, pnictides, and many more have been described. In the course of chemical vapor transport reactions a solid (metallic, covalent, or ionic solids but no molecular crystals) is transferred by a so called transport agent into the gas phase and deposited as well as crystallized at another place. Usually chemical vapor transport takes place at high temperatures (300–1200 °C) in a temperature gradient.

The historical roots of this important preparative method date back to the middle of the 19th century, when several observations on the formation of minerals in nature have been published. In this paper we follow the historical descriptions and development of reactions, which are referred to as chemical vapor transport reactions today.

2 From the Roots to the First Model Concepts

Bunsen, who at this time was a 40 years old professor in Breslau, was the first to describe a naturally occurring vapor transport reaction in 1852.^[1] During the eruption of the volcano Vesuv *Bunsen* noticed the formation of hematite in the presence of hydrogen chloride. Inspired by this observation he

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Harald Schäfer was born on February 10th 1913 in the city of Jena in the German state of Thüringen. For financial reasons he left school in 1929 and joined Schott and Genossen, first as apprentice and later as laboratory technician. During that time he developed the special funnel for fast filtration, still used in many labs today. More important though, he was given the chance to qualify for university studies. Within just three years he completed his studies in chemistry, including thesis work with A. Sieverts at Jena in 1940. The topic of the thesis, determination of boron in glasses, was already suggested by Schäfer himself. In 1948 he got his “Habilitation” in Stuttgart for his work on the System $\text{Fe}_2\text{O}_3\text{-FeOCl-FeCl}_3\text{-H}_2\text{O-HCl}$ and was offered a position in the “Max-Planck-Institut für Metallforschung”. After a short time he moved to Münster as Associate Professor (1953), where he became director of the Inorganic Chemistry Institute and successor to Wilhelm Klemm in 1964. He retired in 1979. Harald Schäfer has received numerous awards for his pioneering work, such as the Alfred Stock Prize of the Gesellschaft Deutscher Chemiker in 1967 and the honorary doctorate from Stuttgart University in 1981.

undertook laboratory experiments on the crystallization of Fe_2O_3 by interaction with chlorine compounds at high temperatures. He called this kind of mineralization “pneumatolysis” and distinguished between a stationary and a pneumatolytic reaction. By the latter a solid can be transferred into the gas-phase at one place and deposited at another. *Bunsen* presented his results on a meeting of the “naturwissenschaftliche Sektion der schlesischen Gesellschaft für vaterländische Kultur”.^[1]

In 1861 *Saint-Claire-Deville* published similar observations in a work with the title “Über die künstliche Nachbildung krystallisierter Mineralien” (On the synthetic replication of crystalline minerals).^[2] In this paper the synthetic principle described by *Bunsen* was applied to other oxides (e.g. Fe_2O_3 , SnO_2 , TiO_2 , MnO , and MgO). For *Saint-Claire-Deville* the observation that a small amount of hydrogen chloride was able to crystallize a much bigger amount of iron(III) oxide was particularly remarkable. It is surprising that *Bunsen*’s paper has not been cited by *Saint-Claire-Deville*.

Troost and *Hautefeuille* have published in 1876 and 1877, respectively, observations on the migration of elemental silicon and aluminum in the presence of halides. However, the authors could not explain the strong temperature dependence of the migration rate.^[3,4] Today it is known that the reason for this phenomenon is the formation of sub-halides at high temperatures. *Troost* und *Hautefeuille* were the first, who focused their experiments not on naturally occurring minerals but rather on synthetic solids. The common subject of these four papers is the description of the crystallization of solids at high temperatures in the presence of reactive gases. All four papers, however, lack a fundamental understanding of the process these phenomena are based on.

In 1890 *Mond*, *Langer*, and *Quincke* published a paper with the title “Action of Carbon Monoxide on Nickel”.^[5] *Mond* and *Quincke* were students of *Bunsen*. Furthermore *Mond* was co-founder of the *Brunner-Mond* Company in Winnington (U.K.). They reported on the volatilization of nickel in the presence of carbon-monoxide. During this reaction gaseous tetracarbonylnickel is formed at 350 °C and decomposed at 450 °C. Thus, re-formation of pure nickel at the higher temperature is achieved. Up to now this reaction is utilized for the industrial purification of nickel. The paper of *Mond*, *Langer*, and *Quincke* is remarkable in several respects. The necessity of a temperature gradient for the volatilization and deposition of the solid by a heterogeneous equilibrium is described for the first time. Furthermore, the authors are the first to postulate a correct reaction equation and, additionally, the first homoleptic metal-carbonyl $\text{Ni}(\text{CO})_4$ is described.

After *Edison*’s patent on the first gas-filled electrical lamp in 1883^[6] there have been several attempts to enhance the light efficiency of these lamps. A significant progress was made in the years 1905 to 1910 through the substitution of the carbon filament by a tungsten-wire. Three years later in 1913 *Langmuir* observed the migration of tungsten from the hot wire to the colder lamp bulb in the presence of water.^[7] By this process the bulb is blackened and thus the life time of the lamp limited. Therefore, he suggested the introduction of chlorine

into the lamps, two years later, in order to induce a migration of tungsten in the opposite direction.^[8] Chlorine should react with the deposited tungsten, forming volatile WCl_6 . This should decompose, re-forming elemental tungsten after migration to the hot wire.

Van Arkel reported in 1923 that pure and crystalline tungsten can be formed by evaporation of WCl_6 followed by decomposition of the vapor at a hot wire, which was heated electrically.^[9] Sometime later this principle was transferred by *van Arkel* and *De Boer* (Philips, Eindhoven) to the formation and purification of titanium, zirconium, hafnium, and thorium.^[10] Utilization of iodine in this process as reactive gas for the transport of the metals is described for the first time. Today this process is often referred to as *iodide process*. In this context a paper by *Geiss* and *v. Liempt* (Philips, Eindhoven) should be mentioned, too.^[11] In 1934 *van Arkel* summarized the state of knowledge in a review article “Die Herstellung hochschmelzender Metalle durch Dissoziation ihrer Verbindungen” (Formation of refractory metals by dissociation of their compounds).^[12] It is noteworthy that the works of *Mond*, *Langer*, and *Quincke*, of *Langmuir* as well as those of *van Arkel* and *De Boer* are strongly related to, if not driven by technical applications.

In the 1930s *Biltz*, *Fischer*, and *Juza* at the University of Hannover were engaged in experiments on the vaporization of solids through the application of reactive gases. Like *Bunsen* the authors used the term pneumatolysis. They report on the chemical reactions involved in this process and the thermochemical background in detail, in a paper titled “Über die pneumatolytische Überführung von Gold mit Chlor” (On the pneumatolytic transfer of gold by means of chlorine).^[13] The absolutely unexpected temperature dependent inversion of the migration direction of gold in a temperature gradient was traced back by the authors to the formation of gaseous gold(III) and gold(I) chloride, respectively. In 1932 *Werner Fischer* was the first to clarify the basic principle of a chemical vapor transport using as example the reaction of alumina with chlorine and hydrogen chloride, respectively (Figure 1).^[14] *Fischer* still used the term pneumatolysis, coined by *Bunsen*.

During the early 1950s the interest in vapor transport was renewed, especially due to unsettled scientific questions, in particular the lack of quantitative understanding. *Schäfer* and *Nickl* picked up in 1953 the paper of *Troost* and *Hautefeuille* (1876),^[3] in which the migration of elemental silicon in presence of gaseous silicon chloride was described in a qualitative manner. The details of the reactions involved in the migration process were determined by measurements and thermodynamic considerations and laid out in detail in the article.^[15] In subsequent papers *Schäfer* described such migration processes quantitatively and developed them to a useful tool for chemists, physicist, crystallographers, and other scientist. Since these fundamental insights the method of chemical vapor transport is strongly related to the name of *Harald Schäfer*.

3 Schäfer’s Model

The groundbreaking paper “Über den Transport des Bodenkörpers im Temperaturgefälle mit Hilfe heterogener Gleichge-

Eine neue Anordnung zur Dampfdruckmessung durch Überführung und ihre Anwendung auf die pneumatolytische Überführung von Al_2O_3 durch HCl bzw. Cl_2

VON WERNER FISCHER und RUDOLF GEWEHR¹⁾

Mit 2 Figuren im Text

1. Einleitung

Leitet man ein Gas über einen festen Stoff, so kann eine Verflüchtigung eintreten, weil der Stoff als solcher verdampft; es kann aber auch ein an sich nicht flüchtiger Stoff vergast werden, indem er unter Reaktion mit dem Gas in chemisch-veränderter Form in die Gasphase übergeht. Unterwirft man den mit den neu entstandenen Stoffen gesättigten Gasstrom aber in bestimmtem Sinne veränderten Bedingungen — z. B. Erniedrigung der Temperatur im Falle negativer Wärmetönung der Verflüchtigungsreaktion — so muß sich bei ausreichender Reaktionsgeschwindigkeit durch Verschiebung des Gleichgewichtes der ursprüngliche feste Stoff wieder abscheiden. Das Erscheinungsbild ist dann also das gleiche wie bei der Sublimation: Transport des Stoffes von Stellen hoher zu solchen tieferer Temperatur; nur ist der Vorgang, der dazu führt, ein anderer, nämlich ein chemischer.

Figure 1. Introduction to the publication by W. Fischer (“A new set-up for vapor pressure measurement by mass transfer and its application to the pneumatolytic transfer of Al_2O_3 by HCl and Cl_2 , respectively”).^[14] He had comprehended brilliantly already almost all aspects of chemical vapor transport reactions.

wichte” (On the transport of a condensed phase in a temperature gradient by means of heterogeneous equilibria) was published in 1956.^[16] In this paper the migration of iron(III) oxide in the presence of hydrogen-chloride in a temperature gradient from 1000 to 800 °C in a closed system (sealed ampoule, Figure 2) is described for the first time. This experimental procedure is used in preparative solid state chemistry up to the present day. By this method a huge variety of quite different solid compounds have been synthesized and crystallized up to now. The materials obtained using this technique enabled a number of novel synthetic applications. In Schäfer’s paper the fundamental relations for the prediction of the transport rate are presented. Additionally, the phrase “chemical vapor transport” was introduced, which is used up to now. It includes

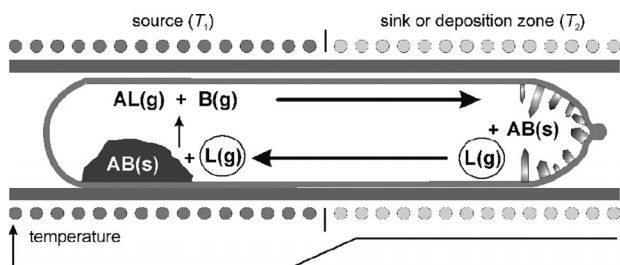


Figure 2. Sketch of a transport ampoule in the temperature gradient of a two-zone tubular furnace. In an exothermic reaction solid AB and the transport agent $L(g)$, which might be regarded as solvent, form the transporting gas species $AL(g)$ and $B(g)$. At the sink, which is for the chosen example at the higher temperature, the reaction is reversed.

the formation of minerals in nature (Pneumatolysis), technical applications such as the *Mond-Langer* process as well as preparative works in the laboratory. Chemical vapor deposition (CVD) is based on the same principles.

Since Schäfer’s fundamental studies sealed ampoules of suitable materials, in most cases silica glass, are used in synthetic solid state chemistry. The reactions take place in a temperature gradient for the duration of a couple of days. The overall reaction is described usually by the following general reaction equation:



It is the characteristic of a chemical vapor transport reaction that only gaseous products are formed off a condensed phase and a gaseous transport agent. In order to account for kinetic effects as well as to allow calculation of transport rates within the whole process commonly three steps are distinguished

- (1) Forward reaction (the condensed phase reacts with the transport agent; first phase-transfer reaction)
- (2) Gas flow
- (3) Back reaction (deposition of the condensed phase under release of the transport agent; nucleation, crystal growth; second phase-transfer reaction)

In order to predict the amount of transported condensed phase it is assumed that the rate-determining step is the gas flow by diffusion. This approximation proves to be true in a pressure range between 10^{-3} and 3 bar. On this basis Schäfer deduced the following relation, which is well known as “Schäfer’s transport equation”. By means of this equation the amount of the transported condensed phase can be calculated in good approximation if the whole process can be described by one (heterogeneous) reaction Equation (2).

$$\dot{n}(A) = \frac{n(A)}{t'} = \frac{i}{j} \cdot \frac{\Delta p(C)}{\Sigma p} \cdot \frac{\bar{T}^{0.75} \cdot q}{s} \cdot 0.6 \cdot 10^{-4} \quad (\text{mol} \cdot \text{h}^{-1}) \quad (2)$$

$\dot{n}(A)$ Transport rate ($\text{mol} \cdot \text{h}^{-1}$)

i, j Stoichiometric coefficients in the transport equation

$\Delta p(C)$ Partial pressure difference of the transport effective species C (bar)

Σp Total pressure (bar)

$\bar{T}$ Mean temperature along the diffusion path (K) (practically $\bar{T}$ can be taken as average of T_1 and T_2)

q Cross section of the diffusion path (cm^2)

s Length of the diffusion path (cm)

t' Duration of the transport experiments (h)

In most cases instead of the factor $0.6 \cdot 10^{-4}$ a value of $1.8 \cdot 10^{-4}$ is given, which found entrance to the literature.^[17] According to recent findings the factor $0.6 \cdot 10^{-4}$ results from a smaller numerical value of the overall diffusion coefficient and corrects a mathematical error.

4 Aim and Prospects of Chemical Vapor Transport Experiments

In the years following the formulation of the transport model, the preparative benefit from chemical vapor transport

reactions was demonstrated by numerous examples. *Schäfer* provided an overview of the theoretical and practical state of knowledge in a monograph, which was published 1962 in German,^[17] in English,^[18] and in Russian.^[19]

The fundamental works of *Schäfer* encouraged other groups to devote their research to this field. *Nitsche* and co-workers used CVT to prepare single crystals of numerous binary and ternary sulfides, selenides, and tellurides. The review articles^[20,21] are dedicated to this field of research. Apart from preparative applications the crystallographer *Nitsche* explored the mechanism of nucleation as well as crystal growth out of the gas phase.^[22] *Emmenegger* studied the transport of oxides in detail.^[23] Furthermore, *Schäfer* and his co-workers were active in this area of research.^[24,25] The large number of publications regarding chemical vapor transport reactions is reflecting the impact of the method on solid state chemistry and materials sciences. In these papers various aspects of CVT experiments are covered:

- (1) Crystal growth for X-ray diffraction experiments
- (2) Crystal growth for studies regarding physical properties
- (3) Deposition of thin films
- (4) Synthesis of novel (metastable) compounds
- (5) Study of phase relationships
- (6) Estimation of thermodynamic data
- (7) Study of gas phase composition
- (8) Further development of model concepts

5 Examples

The following appraisal by *W. Fischer* for the “Transportbuch” written by *Schäfer* seems appropriate as introduction to this section: “Das Verdienst des Autors ist es, die weite Verbreitung der Transportreaktionen erkannt und mit zahlreichen Mitarbeitern die Grundlagen qualitativ und quantitativ geklärt und zahlreiche Beispiele untersucht zu haben. Es ist erstaunlich, wie viele Probleme der verschiedensten Art sich dabei ergeben und sich oft auch gerade mittels Transportreaktionen lösen lassen. So läßt sich z. B. vielfach das verwickelte Nebeneinander mehrerer Gleichgewichte klären, es bieten sich neue Wege zur Ermittlung thermodynamischer Daten, es lassen sich wohl ausgebildete Kristalle von oft hervorragender Reinheit züchten und Stoffe eng begrenzten Existenzbereiches darstellen usw.”^[26] The following classes of substances have been studied and some prominent examples might suffice to underline *Fischer's* statement:

- (1) Elements
- (2) Metal halides
- (3) Binary and multinary oxides
- (4) Oxido-compounds containing complex anions
- (5) Sulfides, selenides, and tellurides
- (6) Chalcogenide halides
- (7) Pnictides
- (8) Intermetallic phases

5.1 Elements

The migration of silver in an oxygen flow was already mentioned in 1877 and confirmed by further research in later

years.^[4,27] Studies regarding the enhanced evaporation rate of the elements iridium, rhodium, and platinum in air point to the same direction.^[28] The first transport of an element that was understood and discussed in detail in the literature was the one of nickel utilizing carbon monoxide. The vast interest in the production of pure metals led to further discoveries in this research area. During the 1920s *van Arkel* and *de Boer* utilized iodine as transport agent thus realizing the vapor transport of different metals. 1934 this subject was summarized.^[12] Investigations show that up to this day about a quarter of all elements can be transported by iodine as transport agent. The thermodynamic background of the metal transport with halogen was studied by *Schäfer*.^[29] In addition, water can enhance the transport rate of certain metals, e.g. tungsten.^[7] Observations of a reversal of the transport direction ($T_2 \rightarrow T_1$ changes to $T_1 \rightarrow T_2$ or vice versa) in dependency of pressure and temperature in the systems Si/I^[30] and Ag/I^[31] were of great importance for the development and understanding of CVT in general. These reversals are caused by a change of the dominating gas species ($\text{SiI}_4/\text{SiI}_2$, $\text{Ag}_3\text{I}_3/\text{AgI}$) at the chosen experimental conditions.

5.2 Metal Halides

Some metal halides show an enhanced volatility in presence of the corresponding halide, which is due to the formation of higher, gaseous halides. This phenomenon was first reported and understood in 1914 studying UCl_4 .^[32] A famous example, which is also prominent in textbooks is the “sublimation” of chromium(III) chloride in a chlorine flow. In the process gaseous CrCl_4 is formed.^[33] By the ingenious application of a so-called three-zone tubular furnace, well controlled syntheses of metal-rich halides (e. g. $\text{Nb}_6\text{Cl}_{14}$,^[33b] Ta_2Cl_5 ^[33c]) have become possible. During the 1970s it was found that many metal halides can react with a second halide, thus forming gaseous complexes. In these molecules several (different) metal atoms are connected by halogen bridges. Due to this mechanism CVT of many metal halides with low volatility became possible using metal halides with high volatility, such as Al_2Cl_6 or Fe_2Cl_6 . Several review articles deal with this phenomenon.^[34,35,36]

5.3 Binary and Multinary Oxides

Up to now hundreds of examples by the transport of oxides are known. Following *Bunsen*,^[1] *Fischer*,^[14] and *Schäfer*, numerous other researchers, including *Emmenegger*,^[22] *Oppermann*,^[36] and *Gruehn*,^[37] explored in detail CVT of oxides.

Chlorinating transport agents (Cl_2 , HCl) are typically used for CVT of metal oxides. Due to their thermodynamic stability iodine is usually not suitable as transport agent. Since 1977 the beneficial properties of tellurium(IV) chloride as transport agent are known. With TeCl_4 the transport of various oxides with rather different oxygen co-existence pressures (thermodynamic stability) was realized. Furthermore tellurium(IV) chloride enables the selective deposition of individual phases in certain systems, in which the oxygen co-existence pressures vary over a wide range and a large number of phases with very

similar compositions is encountered (V/O–TeCl₄, Mn/O–TeCl₄).^[38,39,40] Further examples for the successful crystallization of mixed-valent oxides with neighboring phases of close-lying composition are the Magnéli phases Ti_nO_{2n-1} ($4 \leq x \leq 10$)^[41,42] and of the niobium oxides NbO_x ($2.000 \leq x \leq 2.500$).^[43,44,45]

Crystallization of LaPO₄ was the first example for CVT of an oxide compound containing a complex anion.^[45b] More recently, studies on sulfates and phosphates of the transition metals have been performed, mainly by *Gruehn* and *Glaum*.^[37] For sulfates chlorinating transport agents proved to be the most efficient. For phosphates the utilization of more complex transport agents, like gaseous mixtures phosphorus/iodine or metal phosphide/iodine turned out to be suitable.

5.4 Sulfides, Selenides, and Tellurides

Beginning around 1960 *Nitsche* studied systematically the crystal growth of sulfides and selenides by chemical vapor transport.^[20,46] Iodine is the most common transport agent for this purpose. *Krabbes'* intensive research regarding the transport of sulfides with homogeneity range, such as FeS_x, led to a deeper understanding of the dependence of the transport behavior on the composition of the solid phase. Even small changes in the composition of the solid have a drastic impact on the transport properties.^[47] Currently, compounds such as Cu(In_{1-x}Ga_x)(S_{2-y}Se_y) are of interest for the application in thin film solar cells (CIGS-cells). The physical characteristics of these materials were investigated using single crystals, which have been synthesized via chemical transport reactions.^[48,49,50] Nowadays some selenides and tellurides are in the focus of numerous researchers due to their electrical and thermoelectrical properties. Once again CVT proved to be the method of choice for the preparation of single crystals.^[51]

5.5 Chalcogenide Halides

From the beginning of systematic studies on chemical transport reaction to the present time chalcogenide halides have been playing a key role. The often unexpected formation of solid chalcogenide halides in a reaction of the solid with the transport agent led to numerous investigations on this matter. Especially the discovery of the “auto transport” is closely related to this class of substances. During an auto transport the transport agent is formed through the decomposition of the solid. *Oppermann* provided a survey on this topic.^[52]

5.6 Pnictides

The discovery of III/V-semiconductors such as gallium arsenide triggered numerous works on the crystal growth of these compounds by chemical transport^[48,49,50] and other methods. Fairly new are systematic studies on the transport of phosphides with elucidation of the prevailing heterogeneous and homogeneous reactions.^[37,53,54] Once again iodine is the most common transport agent for these applications. Usually nitrides cannot be synthesized this way but they may be accessible through transport reactions in low-pressure plasma.^[55]

5.7 Intermetallic Phases

Almost 2/3 of all elements are metals. Since most of them form compounds with other metals and metalloids the number of intermetallic phases is extraordinary high. Therefore it is surprising that CVT of this class of substances was only investigated fairly late and by no means in detail. The general applicability of CVT for purification and crystallization of intermetallics has been demonstrated in particular by the studies of *Binnewies* and co-workers.^[48,49,56,57] The fundamental problem in this regard is that all components of the solid have to be transferred into the gas phase and deposited by a single transport agent and at the same experimental conditions (*p*, *T*). Especially components containing an element of group 14 have been studied in detail.^[58,59,60] Since even binary systems may contain many phases, predicting the transport behavior (transport direction, enrichment or depletion of one component in the deposited solid) requires detailed thermodynamic information and application of advanced models (see Section 8).

6 Unusual Reaction Conditions

Kaldis examined transport phenomena, which occur at high temperatures ($\theta > 1200^\circ\text{C}$) (High Temperature Vapor Growth, HTVG).^[61,62] He utilized molybdenum and tungsten as ampoule materials. These ampoules have been heated through high-frequency techniques. *Veprek* studied the transport phenomena in low-pressure plasma^[55] using metal nitrides. Furthermore, the deposition of diamond films from the gas phase shall be mentioned in this context.^[55] The beginning of space travel made it possible to perform experiments at micro-gravity conditions. Experiments with the aim to determine the ratio of diffusion and convection in gas movements have been performed in Soviet and American space stations^[63] by *Oppermann*^[64] and *Wiedemeier*.^[65] These experiments confirmed *Schäfer's* assumption that the diffusion is the leading force of the gas motion. Under the same conditions *Krabbes*^[66] studied the variation of the ¹⁶O/¹⁸O isotope ratio caused by transport phenomena using ¹⁸O enriched iron(III) oxide with chlorine. He traced the phenomena back to different diffusion coefficients of the gaseous oxygen species.

Nowadays, hydrothermal (or more general solvothermal) syntheses count as chemical transport reactions as well. An overview is given by *Rabenau*.^[67]

7 Gas-Phase Chemistry

Due to the advances of chemical vapor transport and its establishment as a preparative method in solid state chemistry the interest in qualitative and quantitative description of the gas phase composition increased. For further investigations the classical measurement methods as described in the book of *Kubaschewski*^[68] (transpiration methods, total pressure measurements, *Knudsen* method) as well as modern spectroscopic methods (UV/Vis, Raman-spectroscopy, mass spectroscopy) have been applied. Thus the knowledge of gas phase chemistry at high temperatures, related to CVT was enormously im-

proved and extended. Nowadays, quantum chemical methods are used increasingly in order to provide predictions about structure and stability of possibly occurring gaseous molecules (see chapters 11 and 12 in Ref. [48] and Ref. [49]).

8 Conceptual Progress Since 1970

Until the late 1960s more and more experimental observations in publications by many different authors had demonstrated the validity of *Schäfers* approach to the thermodynamic understanding of chemical vapor transport experiments. Simultaneously, chemical vapor transport developed to one of the most widely used techniques for the crystallization and purification of solids on a laboratory scale. The rather simple set up for experiments in sealed ampoules (Figure 2) and the underlying theory rooted soundly in thermodynamics allow generally good control of the experimental conditions and high reproducibility of the results. Furthermore, chemical vapor transport frequently provided experimental evidence for hitherto unknown gaseous molecules, stable only at high temperatures in co-existence (see Section 7) with a condensed phase. Last but not least, CVT experiments provided the basis for the estimation of thermodynamic data for condensed phases and gaseous molecules (e.g. $\text{MoO}_2\text{I}_2(\text{g})$ ^[69] and $\text{GaAs}_5(\text{g})$ ^[70]).

It was then, that experimental observations became increasingly obvious, which failed to comply with *Schäfers* rather simple model of chemical vapor transport (transport by diffusion between two thermodynamic equilibrium spaces containing two identical single-phase solids; analysis of a single heterogeneous equilibrium, as depicted in Section 3). These deviations from the simple model might be summarized by the following observations

(1) Chemically complicated transport systems may involve more than just a single heterogeneous reaction as well as several homogeneous gas-phase equilibria.

(2) The temperature dependency of heterogeneous reactions, which change the number of gas molecules will in a temperature gradient not only cause mass flow by diffusion due to partial pressure differences but will also lead to a directed flow (*Stefan* flow).^[48,49]

(3) For chemically complicated transport systems, in particular for those with several co-existing condensed phases, equilibrium calculations following the standard K_p procedure^[48,49,71] become increasingly difficult.

(4) Equilibration between the starting solid and the gaseous transport agent shifts the composition of a solid with homogeneity range ("incongruent dissolution of a solid in the gas phase"); for compounds without homogeneity range such reactions might lead to formation of a second (third,...) condensed equilibrium phase.

(5) In particular for preparative purposes the question arose, how the composition of transported crystals with homogeneity range can be controlled.

(6) Transport experiments with solid solutions or multi-phase solids showed that for these cases the mass transfer might not be independent of time ("non-stationary" transport behavior).

(7) For multinary compounds as well as for those with homogeneity range de-mixing with sequential CVT of the components was observed.

(8) Occasionally, CVT experiments led to deposition of thermodynamically metastable phases (e.g. CoP_2 instead of two-phase mixtures CoP/CoP_3).^[72]

8.1 Solubility of a Condensed Phase in the Gas-Phase

It was *Schäfer* himself, who led the way in how to understand experiments, where CVT of a solid cannot be completely described by *just one* reaction, since a more complex gas phase is formed. For these chemically more complicated cases several unique equilibrium reactions have to be considered. He coined the expression of the solubility of a solid in the gas phase.^[73] This solubility λ is defined according to Equation (3) as the ratio of the amounts of dissolved solid *A* in gaseous solvent *L*. The amounts can be expressed either by the balance of molar numbers n^* or partial pressures p^* .

$$\lambda(A) = \frac{n^*(A)}{n^*(L)} = \frac{p^*(A)}{p^*(L)} \quad (3)$$

One example, the solubility of iron in gaseous iodine, Equation (4) should suffice for visualization.

$$\lambda(\text{Fe}) = \frac{p^*(\text{Fe})}{p^*(\text{I})} = \frac{p(\text{FeI}_2) + 2p(\text{Fe}_2\text{I}_4)}{p(\text{I}) + 2p(\text{I}_2) + 2p(\text{FeI}_2) + 4p(\text{Fe}_2\text{I}_4)} \quad (4)$$

Substitution of Δp in Equation (2) by Equation (5) allows calculation of transport rates based on solubility differences between source and sink instead of using the partial pressure difference for a single gas molecule. Use of the solubility difference is even more recommended, since it accounts already for possible *Stefan* flow, as has been demonstrated by *Schäfer*.^[73]

$$\Delta p = \Delta\lambda(A) \cdot \overline{p^*(L)} = \left(\lambda(A)_{T_{\text{source}}} - \lambda(A)_{T_{\text{sink}}} \right) \cdot \overline{p^*(L)} \quad (5)$$

8.2 Equilibrium Calculations by the $G_{\min}$ Method

The calculation of equilibrium partial pressures and the related determination of the condensed equilibrium phases applying the K_p method^[71] become for more complicated systems rather cumbersome, if not impossible. Therefore, *White*, *Johnson*, and *Dantzig* suggested already in 1958 equilibrium calculations based on the minimization of the Gibbs Free Energy of a system ($G_{\min}$ method).^[74] It was only in 1971 though, that a computer program, developed by *Eriksson*, allowed such calculations to be performed routinely.^[75] Details on the method can be found in literature.^[48,49,76] The programming approach used by *Eriksson* is nowadays adopted in all software aiming at the calculation of phase diagrams ("CALPHAD"^[77]). It is also implemented in the two computer programs TRAGMIN^[78] and CVTRANS,^[79] which are currently in use for the modeling of CVT experiments.

8.3 CVT of Phases with Homogeneity Range

Thermodynamic description and modeling of transport systems get increasingly complicated if the transported compound shows a homogeneity range $AB_{x \pm \delta}$ or the transport occurs in a system with several coexisting condensed phases, e.g. AB_y and AB_z . For all of these cases congruent dissolution of the condensed phase in the gas phase becomes rather unlikely. Incongruent dissolution should be anticipated. This case is characterized by different ratios of the molar numbers for the components (elements) in the source solid and the corresponding gas phase. Such a behavior is comparable to the peritectic melting of a solid and the compositional shift that accompanies the re-formation of a solid from this melt upon cooling. The compositions of melt and solid are different. As an immediate consequence of incongruent dissolution in the gas-phase, the composition of the solid at the sink (deposition) side, expressed by the ratio $(n(A)/n(B))$ in AB_x at $T(\text{sink})$ is no longer identical to the ratio of the balance pressures $p^*(A)/p^*(B)$ of the components A and B at the sink. In the same way the ratio of the components of the deposited condensed phase does not need to be identical to that of the dissolved phase. Thus, in the sink a phase might be deposited which is different to the one present at the source.

CVT of $\text{TiS}_{2-\delta}$ by iodine as transport agent was one of the first examples for the aforementioned transport behavior.^[80] In a temperature gradient $950 \rightarrow 850^\circ\text{C}$, starting from a source solid with higher titanium content, in the sink deposition of a sulfur-enriched solid is found ($\delta_{\text{source}} > \delta_{\text{sink}}$) (Figure 3). For a given starting composition and temperature gradient, the composition of the deposited crystals can be read directly from the barogram of the binary system Ti/S .^[80,48,49] The equilibrium pressure of sulfur $p(\text{S}_2)$ at the source determines the composi-

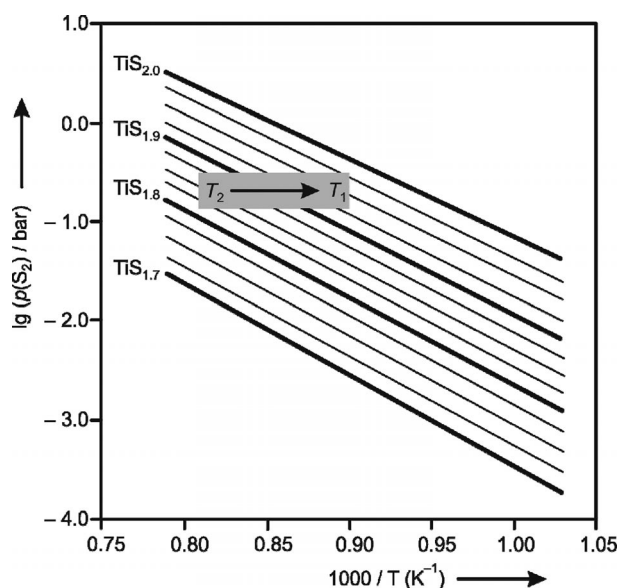


Figure 3. Barogram of the system Ti/S showing the co-existence pressures in the compositional range $\text{TiS}_{2-\delta}$. The arrow indicates the compositional shift observed in chemical vapor transport experiments in a temperature gradient $950 \rightarrow 850^\circ\text{C}$; graphics according to Ref. [80] and Ref. [49].

tion of $\text{TiS}_{2-\delta}$ at the sink. Obviously, with the depletion of sulfur in the source solid the composition of the crystals deposited at the sink must change. In ampoule experiments, quasi-stationary transport with deposition of $\text{TiS}_{2-\delta}$ within a very narrow compositional range can be achieved when using a large amount of starting material and finishing the experiment after only a rather small amount of crystals were deposited.

A general analysis of the problem has been given by *Krabbes*, *Oppermann*, and *Wolf*, by introducing the extended transport model.^[81–84] They realized that the composition of a binary compound AB_x deposited at the sink is determined by the ratio of the fluxes $J(B)$ and $J(A)$ from the source to the sink (Equation (6)).

$$\left(\frac{n(B)}{n(A)} \right)_{T_{\text{sink}}} = \left(\frac{\text{flux}(B)}{\text{flux}(A)} \right)_{T_{\text{source}} \rightarrow T_{\text{sink}}} = \frac{J(B)}{J(A)} = x_{\text{sink}} \quad (6)$$

By linking the fluxes $J(B)$ and $J(A)$ of the individual components of the system and assuming that the net flux of the transport agent X will vanish, $J(X) = 0$, the stationarity relation ε (Equation (7)) was derived^[75,43,44]. Thus, it was shown, how equilibrium calculations for source and sink region have to be linked to each other, without introducing some unjustified restrictions concerning the composition of the equilibrium solids. This so-called *extended transport model* is implemented in the computer program TRAGMIN,^[72] which is particular useful for predicting the composition of crystals with homogeneity range and multiphasic mixtures deposited at the sink in a quasi-stationary transport (e.g.: CVT of iron sulfides FeS_x ^[75,76,77] and vanadium oxides $\text{V}_n\text{O}_{2n-1}$.^[38,39,40]

$$\left(\frac{p^*(B) - x_{\text{sink}} \cdot p^*(A)}{p(X)} \right)_{\text{source}} = \left(\frac{p^*(B) - x_{\text{sink}} \cdot p^*(A)}{p^*(X)} \right)_{\text{sink}} = \varepsilon \quad (7)$$

In addition to calculation of equilibrium partial pressures and condensed phases (Figure 4) the extended transport model offers further information on experimental realization and theoretical understanding of chemical vapor transport. Thus, calculation of the transport efficiency of individual gas species and deduction of the prevailing transport reaction(s) become possible. By using the *extended transport model* elucidation of the various reactions involved in the transport of oxides VO_x by TeCl_4 was possible. These calculations show a change in the prevailing tellurium-containing gas species from Te_2 to TeCl_2 (and TeOCl_2) with increasing oxygen content of the solid (Figure 5). Thorough analysis of the effective flow from the source to the sink of the various gas species led to the identification of the following reactions (Equation (8), Equation (9), Equation (10), and Equation (11)) to be responsible

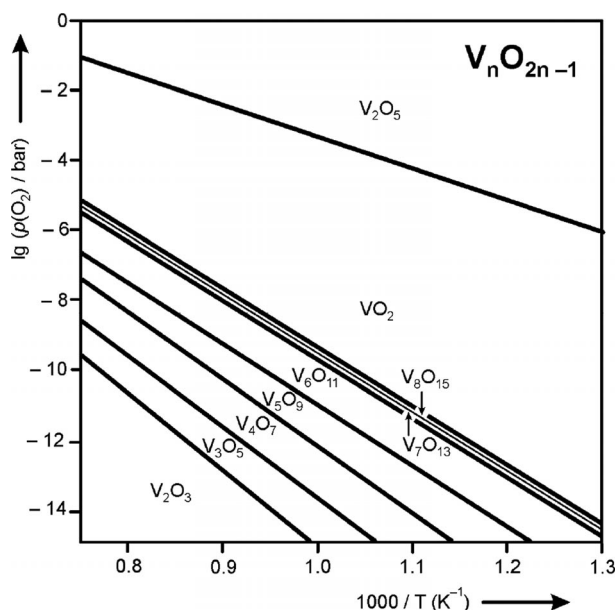


Figure 4. Barogram of the vanadium/oxygen system according to *Oppermann* and co-workers.^[38,40]

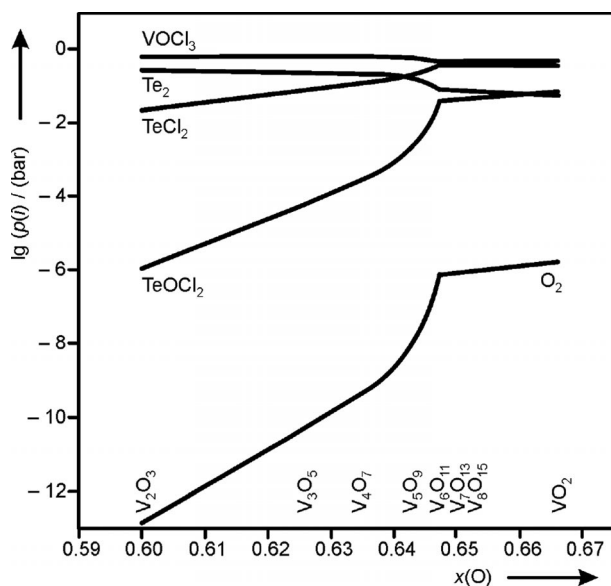
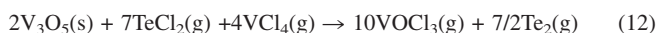
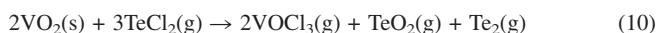
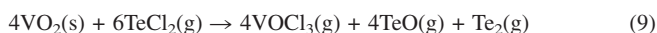
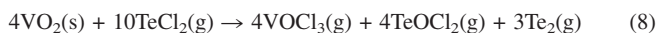


Figure 5. Gas-phase composition obtained for the dissolution of V_2O_3 , the Magnéli phases, and VO_2 by $TeCl_4$ at 1000 °C. Starting amounts: 0.01 mol VO_x ; 5×10^{-3} mol l^{-1} $TeCl_4$.^[49]

for the transport of VO_2 . Accordingly, a combination of reactions (Equation (12), Equation (13), and Equation (14)) describes the CVT of V_3O_5 :^[38–40,49]



8.4 Non-Stationary Transport Experiments

Almost simultaneously to the extended transport model the model of co-operative transport was introduced by *Gruehn* and *Schweizer*.^[45,85] Both models include equilibrium calculations for spatially separated source and sink regions with an effective mass flow between them. In contrast to the quasi-stationary treatment of the extended transport model the co-operative transport model attempts to describe the whole (“non-stationary”) course of a transport experiment. By repeated source and sink calculations the time-dependent mass transfer from the source to the sink is simulated. Thus, the sequential, sometimes simultaneous, migration of multi-phase solids to the sink can be analyzed according to the underlying thermodynamics. Over a period of 20 years many examples for non-stationary transport experiments have been analyzed in considerable detail by *Gruehn* and co-workers. Subsequently, just a few are presented:

(1) Deposition of $Nb_{12}O_{29}$ before NbO_2 starting from a two-phase source solid $Nb_{12}O_{29}/NbO_2$ (transport agent: $NbCl_5$, 1125 → 1025 °C).^[45]

(2) Deposition of WO_2 before $W_{18}O_{49}$ starting from a two-phase source solid $W_{18}O_{49}/WO_2$ (transport agent: HgI_2 , 1060 → 980 °C).^[86]

(3) Deposition of $CrCl_3$ before $CrOCl$ starting from a source solid $CrOCl$ (transport agent: Cl_2 , 600 → 500 °C);^[87] deposition of $CrCl_3$ before Cr_2O_3 starting from a source solid $CrOCl$ (transport agent: Cl_2 , 900 → 800 °C).^[88]

(4) Formation and sequential transport of several condensed phases when transporting $NiSO_4$ by addition $PbCl_2$ (850 → 750 °C).^[89]

The time dependent behavior of the aforementioned examples has been studied experimentally using a so-called *transport balance*.^[89] A sketch of the experimental set up is given in Figure 6.

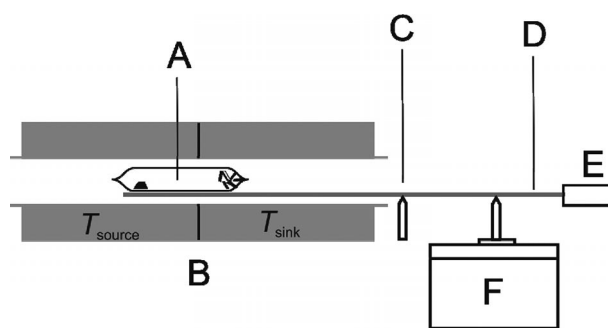


Figure 6. Sketch of a the transport balance as introduced by *Gruehn* and co-workers.^[89] **A** transport ampoule, **B** tubular two-zone furnace, **C** and **D** beam balance with sapphire pivot, **E** counter weight, **F** digital micro-balance.

In particular the course of the transport of $NiSO_4$ with $PbCl_2$ as transport additive (not transport agent!) is quite enlightening and shows the potential of both, the experimental set up (transport balance) and the theoretical treatment (co-operative transport model) for analyzing complex transport systems (Fig-

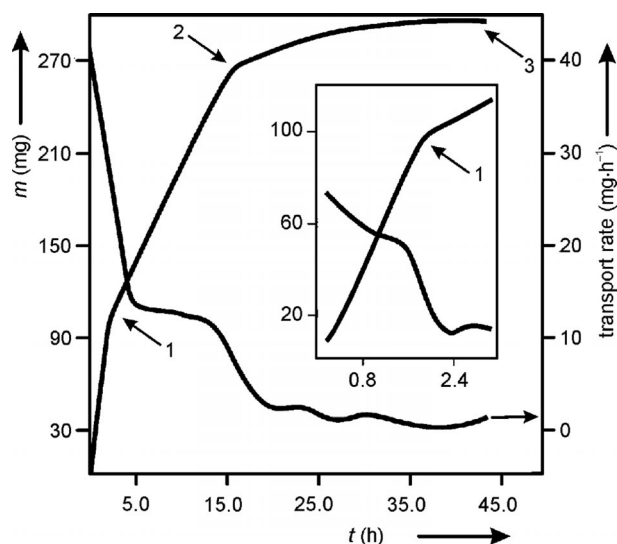
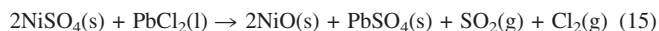


Figure 7. Curves for deposited mass vs. time and transport rate vs. time as registered by a transport balance for the chemical vapor transport in the system $\text{PbCl}_2/\text{PbSO}_4/\text{NiSO}_4$.^[89] The inset shows the steady for the deposition of PbSO_4 (accompanied by small amounts of PbCl_2) at the beginning of the experiment (until point 1). Between points 1 and 2 transport of NiSO_4 takes place. Mass changes after point 2 are caused by migration of solid within the sink due to small temperature gradient in this region; termination of the experiment at point 3.

ure 7). The authors could show, that in a pre-reaction (equilibrium) (Equation (15)) PbSO_4 and NiO are formed besides NiSO_4 and a surplus of the transport additive. Out of this set of equilibrium condensed phases in the source in a first quasi-stationary state distillation of PbCl_2 takes place (Equation (16)), followed by the transport of PbSO_4 , according to Equation (17). NiSO_4 is eventually transported during a third quasi-stationary state (Equation (18)). It should be noted, that transition from one quasi-stationary state to the next will always be accompanied (caused) by a change of the gas-phase composition.



9 Conclusions

From first observations 150 years ago, chemical vapor transport has been developed during the last six decades into an indispensable method for crystallization and purification of solids on a laboratory scale. *Harald Schäfer* deserves the merits for providing understanding of the underlying principles, in particular the influence of thermodynamics. Thus, he showed that there is much more to gain from CVT experiments than nice crystals. By careful and meticulous analysis of the results

of chemical vapour transport experiments information on new gaseous molecules, thermodynamic data of involved solids and gas-species as well as hints on possible kinetic effects related to nucleation and growth of crystals can be obtained. Even in those solid state syntheses where no migration of a solid in a temperature gradient can be achieved, knowledge of the aforementioned principles and reactions will allow a much more reasonable selection of mineralizers and reaction conditions.

The controlled synthesis of tailor-made solids, in most cases compounds showing a homogeneity range, demands for detailed understanding of the more complex principles elucidated only during the last decades. Even for syntheses in open systems (“flow tubes”) modeling of the chemical transport reactions by the described theoretical approaches and computer programs will turn out to be most fruitful. Eventually, one might predict, that the increased demand for re-cycling of (precious) metals and their compounds will lead to a renaissance of chemical vapor transport reactions.

Supporting Information (see footnote on the first page of this article): A complete list of publications by *Harald Schäfer* can be found in the literature.^[90] Lists of publications by *Reginald Gruehn* and *Heinrich Oppermann* are deposited.

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- When a gas passes over a solid a volatilization can occur because the solid evaporates. Solids with very low (autogenous) vapor pressure can be volatilized, by a chemical reaction between the solid and the gas. The gas phase contains the solid in chemically modified form. If one changes the conditions of the so formed equilibrium gas, e.g. by lowering the temperature in the case of an endothermic volatilization reaction, from a thermodynamic viewpoint the solid must be reconverted due to the modified equilibrium position. The phenomenon is just the same as in case of a sublimation reaction: the solid migrates from higher to lower temperature, but the reason for this migration is a chemical reaction.

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